

Table 3-5. Summary of the Coverage of Student Outcomes in Chemical Engineering Courses.

Student Outcomes		Number of courses at the indicated level of coverage			
#	Description	4 Topic is main focus of course, grade largely dependent on demonstrated knowledge;	3 Course grade partially dependent on demonstrated knowledge on this topic;	2 Topic introduced in course, course grade not dependent on demonstrated knowledge;	1 Topic not covered specifically in course.
1	An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	18	4	3	1
2	An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	9	6	4	6
3	An ability to communicate effectively with a range of audiences.	7	11	4	4
4	An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	6	7	4	7
5	An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	8	5	2	9
6	An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	5	5	4	10
7	An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.	8	8	4	6